

Member ID: _____

Time: _____

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DEVICE CONFIGURATION & TROUBLESHOOTING (305)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice: Identify the letter of the choice that *best* completes the statement or answers the question.

1. The default file system used in Windows 10 is _____.
 - a. NTFS
 - b. FAT64
 - c. FAT32
 - d. NTFS64
2. What tool will show you the full network path IP addresses from one device to another?
 - a. ping
 - b. ifconfig
 - c. tracert
 - d. ipconfig
3. What is the Windows command line utility that views and manages drives and partitions?
 - a. chkdsk
 - b. list disk
 - c. tracert
 - d. diskpart
4. What Linux command will change the directory you're viewing?
 - a. mkdir
 - b. cd
 - c. chmod
 - d. ls
5. What must you do to make a new drive useable within Windows?
 - a. Run a defrag
 - b. Format the drive in File Explorer
 - c. Format the drive in Disk Management
 - d. Run chkdisk
6. What should be your first step to troubleshoot Windows after installing a new device driver and encountering a no boot situation on reboot?
 - a. Boot to Safe Mode with Networking
 - b. Boot to Windows media
 - c. Boot to Safe Mode
 - d. Boot to Debug Mode

7. What Windows command scans the file system on a disk drive and analyzes the integrity of the files, file system, and file metadata on the drive?
 - a. verifier /standard /all
 - b. compact vdisk
 - c. regsvr32 /u
 - d. chkdsk /scan
8. You are working as an IT administrator for a small company. A salesman has a laptop that requires a memory upgrade in order to run the software needed for their job role. What type of memory would you most likely need to install on their laptop?
 - a. RDIMM
 - b. SODIMM
 - c. NVDIMM
 - d. UDIMM
9. What feature is needed on a switch for a wireless access point to bypass using a secondary power connection?
 - a. QoS
 - b. IPv6
 - c. TCP/IP
 - d. PoE
10. Which of the following is a Class A subnet mask?
 - a. 255.255.0.0
 - b. 255.0.0.0
 - c. 255.255.255.0
 - d. 255.255.254.0
11. Every time your computer boots it takes an excessive amount of time. You notice that it attempts to PXE boot before loading Windows. What could do you to speed up the boot time process?
 - a. Update BIOS
 - b. Load the one time boot menu and select your device
 - c. Edit your BIOS boot settings
 - d. Change boot mode from BIOS to UEFI
12. As a part of troubleshooting a printer that will not print documents from a system, you wish to restart the service in a Windows PC. Where would you find this option?
 - a. Start > Settings > Devices > Printers & Scanners > Select the printer
 - b. Start > Settings > Apps > Print Spooler
 - c. In the taskbar search, type Services and locate Print Spooler in the list of services
 - d. Open Windows PowerShell and run the command '*Set-Service -Name Spooler -StartupType Disabled*'

13. You have been hired to assist designing and setting up a network that runs through a wired connection in a business. The building has a higher than normal amount of EMI and is approx. 80 meters from end to end. What would be the *most* cost effective and *most* ideal cable option to run through the business?
 - a. STP Cat-8
 - b. Coaxial
 - c. UTP Cat-6
 - d. STP Cat-5e
14. What Windows command would you use to locate and consolidate fragmented files to increase system performance?
 - a. sfc
 - b. chkdsk
 - c. defrag
 - d. diskpart
15. During troubleshooting a Windows no boot situation, you need to scan the integrity of all protected system files, and repair files with problems when possible, to attempt to restore the OS. What command would be appropriate to use?
 - a. SFC /scannow
 - b. SFC /verifyonly
 - c. DISM /Online /Cleanup-Image /RestoreHealth
 - d. DISM /Online /Cleanup-Image /ScanHealth
16. You are an administrator of a Linux environment. There is a driver update file you wish to rename in order to more easily identify. What command would be appropriate to perform this task?
 - a. rn oldfile.bin newfile.bin
 - b. mv oldfile.bin newfile.bin
 - c. ls oldfile.bin newfile.bin
 - d. rename 's/oldfile.bin/' *.bin
17. After power off, what is the first thing you should do before opening a portable computer for repairs?
 - a. Connect an anti-static strap to bare metal
 - b. Remove the battery
 - c. Hold the power button down for 30 seconds to drain power
 - d. Place parts on an anti-static mat
18. Which one of these is *not* a primary function of the Basic Input/Output System?
 - a. Scans for a boot device and hands off to the best OS available
 - b. Checks the hardware health for any errors
 - c. Initializes firmware for the hardware devices
 - d. Configures storage raid arrays.

19. Which partition type supports drives larger than 2 TB?
 - a. GPT
 - b. FAT32
 - c. MBR
 - d. FAT
20. Which of these devices is generally considered a hot swappable device?
 - a. Small outline dual in-line memory module
 - b. Internal network interface card
 - c. External universal serial bus solid state drive
 - d. Graphics processing unit
21. Which of these would *not* meet the minimum system requirements to install Windows 10 32-bit operating system?
 - a. 1 GHz CPU
 - b. 800 x 600 display
 - c. 15 GB hard disk space
 - d. 1 GB of ram
22. Every time you reboot your system you notice that the time is incorrect. You reset the time, but every reboot it shows incorrectly again and has to be reset. What is the *most* likely suspect?
 - a. Outdated drivers
 - b. Corrupted Windows installation
 - c. Failed CMOS battery
 - d. Failing hard drive
23. You keep commonly failed component shelf spare parts in storage for quick and easy repair of the systems you administrate. What is the best way to store these components in order to protect them? (*Choose the best answer*)
 - a. On top of an anti-electrostatic bag
 - b. In a non-air conditioned warehouse
 - c. Inside of an anti-electrostatic bag
 - d. Inside an anti-electrostatic bag in a climate controlled area
24. You are setting up a new end-user system for an employee at your company. You connect the system to the network and receive an automatically assigned IP address for network communication. What service is responsible for this IP assignment?
 - a. DHCP
 - b. IPv4
 - c. DNS
 - d. TFTP
25. A printer in your office is printing out pages with streaks and smudges making documents unclear and unusable. What is the most likely cause of this?
 - a. Empty toner cartridge
 - b. Overheating of the printer
 - c. Failing printer drum
 - d. Jammed paper

26. An employee brings you their laptop because it has experienced a BSOD. What file would you want to check first to start troubleshooting?
- Audit log
 - Page file
 - Setup log
 - Dump log
27. In a modern day Linux distribution, what is the most default file system?
- Ext3
 - Ext4
 - Ext64
 - Ext32
28. On your computer, you have a C drive for Windows files and a D drive for other data. After a drive failure on the system you discover you lost all data on both the C and D drives. What is the most likely reason for this?
- The drives were part of a RAID-0
 - Both drives were partitions on the same disk
 - Both drives failed at the same time
 - Radiation exposure corrupted the data
29. You are troubleshooting a network problem on a Linux system. What command is needed to view the IP address of the system?
- ipconfig
 - ping
 - netsh
 - ifconfig
30. You need to update a Windows system to the newest driver for its network card. What is the best place to find the newest available driver version?
- Google
 - Windows installation disk
 - OEM tech support
 - Manufacturer's website
31. You are troubleshooting a system failure. There is no light on the power button when pressed, and no lights on the network card. What kind of troubleshooting path should you follow?
- No power
 - No boot
 - No POST
 - No video

32. You have obtained a new external hard drive that will migrate between both Windows 10 and Ubuntu systems. What file system should be selected to format the new drive?
- APFS
 - Ext4
 - HFS+
 - NTFS
33. What is another name for Parallel ATA?
- IDE
 - SAS
 - SATA
 - NVMe
34. You have been given a system that does not boot stating that no boot device is available. What is an easy first step to troubleshoot?
- Replace the system drive
 - Change the boot mode from BIOS to UEFI
 - Reformat the drive and reinstall the OS
 - Replace the system power cord
35. You are engaged by an end user complaining that their system performance is very slow. What would be the best tool to analyze what programs are using the most system resources?
- Event Viewer
 - Services
 - Task Manager
 - Control Panel
36. You need to add a manufacturer's website to the trusted sites for your local system for obtaining driver downloads. Where would you go in a Windows OS to do this?
- Control Panel > Internet Options > Privacy > Trusted Sites
 - Control Panel > Network and Sharing Center > Security > Trusted Sites
 - Control Panel > Network and Sharing Center > Privacy > Trusted Sites
 - Control Panel > Internet Properties > Security > Trusted Sites
37. Translate the decimal value of 220 into Hexadecimal.
- 0x220
 - 0xDC
 - 0xB3
 - 0x5E
38. Which of the following statements best describes a Simple Layer-2 Network Switch?
- Redirects traffic from one network to another
 - Repeats all traffic sent to one port out of all other ports
 - Directs traffic by examining its MAC address and sending to the port where that addresses owner is located
 - Directs traffic by examining its IP address and sending to the port where that addresses owner is located

39. After completing a Windows installation, navigating to the Device Manager reveals several components have yellow triangles with exclamation marks on them. Based on this, what would need completed prior to using the system further?
- Install another version of Windows
 - Install a virus scanner
 - Remove all marked devices from inventory
 - Install device drivers
40. Convert the number 226 into 128-bit binary.
- 11000001
 - 11100010
 - 00100101
 - 00101010
41. Keyboard shortcut for "Cut" in most word processor application is which of the following combinations?
- Ctrl+X
 - Ctrl+C
 - Ctrl+Alt+Del
 - Ctrl+Z
42. For what would we use a loopback cable?
- Connecting an end user system to the network switch
 - Test physical ports to identify network issues
 - Automatic configuration of a physical port
 - Establishing a serial connection to a network device
43. What is used as a kind of address book to translate a web address into and IP address and back again?
- DHCP
 - TCP/UDP
 - TFTP
 - DNS
44. Which process should be done regularly to keep a system running smoothly with as few unplanned outages as possible?
- Update the drivers, firmware, and OS
 - Upgrade the RAM
 - Reimage the OS
 - Reinstall applications
45. What Windows feature assigns an IP address to network ports when no DHCP is available?
- DNS
 - NETBIOS
 - Wireshark
 - APIPA

46. Which of these is a valid OSI layer-2 address?
- a. A4-BB-6D-EF-A2-2E
 - b. 2001:0bb8:8500:0000:0000:8a2e:0370:7221
 - c. 192.168.0.254
 - d. fe80::35bd:6f2a:bc6d:e31d%23
47. All 64-bit operating systems can run 32-bit applications. True or False?
- a. True
 - b. False
48. What tool is used to test if an IP address is reachable within a network?
- a. SSH
 - b. Telnet
 - c. Ping
 - d. Packet Tracer
49. CPUs that use a grid of flat metal contact that connect to blunt pins on the system board are using what type of socket?
- a. Zero Insertion Force Array
 - b. Land Grid Array
 - c. Ball Grid Array
 - d. Flat Force Array
50. When configuring permissions on a folder in a network for users to access, the permissions do not apply to the subfolders within.
- a. True
 - b. False